

TABLE I: Comparison of quantitative results on UHD-LL. UHdRes achieves the highest PSNR with a relatively small number of parameters.

| Method          | Source    | Params↓      | PSNR↑         | SSIM↑         | LPIPS↓       |
|-----------------|-----------|--------------|---------------|---------------|--------------|
| IFT [9]         | ICCV’21   | 11.56M       | 21.960        | 0.8700        | 0.324        |
| SNR-Aware [10]  | CVPR’22   | 40.08M       | 22.720        | 0.8770        | 0.304        |
| Restormer [7]   | CVPR’22   | 26.1M        | 21.536        | 0.8437        | 0.361        |
| Uformer [6]     | CVPR’22   | 20.6M        | 21.303        | 0.8233        | -            |
| LLFormer [32]   | AAAI’23   | 13.2M        | 22.790        | 0.8530        | 0.264        |
| DiffLL [18]     | TOG’23    | 17.29M       | 21.360        | 0.8720        | 0.239        |
| UHDFour [15]    | ICLR’23   | 17.5M        | 26.226        | 0.9000        | 0.239        |
| UHdformer [34]  | AAAI’24   | <b>0.34M</b> | 27.113        | 0.9271        | 0.245        |
| LMAR [44]       | CVPR’24   | 1.97M        | 26.270        | 0.9196        | 0.225        |
| Wave-Mamba [14] | ACM MM’24 | 1.26M        | 27.350        | 0.9130        | <b>0.185</b> |
| MambaIR [11]    | ECCV’24   | 2.01M        | 24.127        | 0.8965        | 0.284        |
| AdaIR [16]      | ICLR’25   | 1.05M        | 25.669        | 0.9014        | 0.268        |
| UHDDIP [45]     | TCSVT’25  | 0.81M        | 26.749        | 0.9281        | <b>0.208</b> |
| ERR [17]        | CVPR’25   | 1.13M        | <b>27.570</b> | <b>0.9320</b> | 0.214        |
| UHdRes          | Ours      | <b>0.40M</b> | <b>27.693</b> | <b>0.9311</b> | 0.231        |